

Jupyter Notebook Execution Report

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Cell 1: ■ Markdown

TP 3

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Cell 2: ■ Code

```
# import libraries

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.metrics import adjusted_rand_score
```

Cell 3: ■ Code

```
documents = ["This ltitle kitty came to play when I was eating at a restaurant.",
"Merley has the bets squooshy kitten belly.",
"Google Translate app is incredible.",
"If you open 100 tab in google you get smiley face.",
"Best cat photo I've ever taken.",
"Climbing ninja cat.",
"Impressed with google map feedback.",
"Key promoter extension for Google Chrome."
]
```

Cell 4: ■ Code

```
vectorizer = TfidfVectorizer(stop_words='english')
```

Cell 5: ■ Code

```
x = vectorizer.fit_transform(documents)
```

Cell 6: ■ Code

```
print(vectorizer.vocabulary_)
```

Output:

```
{'ltitle': 19, 'kitty': 18, 'came': 5, 'play': 25, 'eating': 9, 'restaurant': 27, 'merley': 21, '
```

Cell 7: ■ Code

```
print(vectorizer.get_feature_names_out())
```

Output:

```
[ '100' 'app' 'belly' 'best' 'bets' 'came' 'cat' 'chrome' 'climbing'
  'eating' 'extension' 'face' 'feedback' 'google' 'impressed' 'incredible'
  'key' 'kitten' 'kitty' 'ltitle' 'map' 'merley' 'ninja' 'open' 'photo'
  'play' 'promoter' 'restaurant' 'smiley' 'squooшы' 'tab' 'taken'
  'translate' 've']
```

Cell 8: ■ Code

```
true_k = 2
model = KMeans(n_clusters=true_k, init="k-means++", max_iter=100, n_init=1)
model.fit(x)
```

Output:

```
KMeans(max_iter=100, n_clusters=2, n_init=1)
```

Cell 9: ■ Code

```
print(model.labels_)
```

Output:

$$[1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0]$$

Cell 10: ■ Code

```
print("Prediction")
```

```
y = vectorizer.transform(["chrome browser to open"])
prediction = model.predict(y)
print(prediction)
```

Output:

```
Prediction
[0]
```

Cell 11: ■ Code

```
y = vectorizer.transform(["My cat is hungry."])
prediction = model.predict(y)
print(prediction)
```

Output:

```
[1]
```

Cell 12: ■ Code

```
print("Top terms per cluster:")
order_centroids = model.cluster_centers_.argsort()[:,::-1]
terms = vectorizer.get_feature_names_out()
for i in range(true_k):
    print("Cluster %d: " %i),
    for ind in order_centroids[i,:10]:
        print(" %s" % terms[ind])
    print
```

Output:

```
Top terms per cluster:
Cluster 0:
google
translate
feedback
map
incredible
impressed
```

```
app
extension
key
promoter
Cluster 1:
cat
ninja
climbing
ve
photo
taken
best
belly
merley
squoooshy
<built-in function print>
```

Cell 13: ■ Code

```
import pandas as pd

results = []

# we only have 8 documents, can't have k > 8.

for k in range(3, 9):
    model = KMeans(
        n_clusters=k,
        init="k-means++",
        max_iter=100,
        n_init=1,
        random_state=42
    )
    model.fit(x)

    order_centroids = model.cluster_centers_.argsort()[:, :-1]

    for cluster_id in range(k):
```

```

top_terms = [terms[index] for index in order_centroids[cluster_id, :10]]
results.append({
    "k": k,
    "Cluster": cluster_id,
    "Top terms": ", ".join(top_terms)
})

results_table = pd.DataFrame(results)
results_table

```

Output:

	k	Cluster	Top terms
0	3	0	google, translate, feedback, map, incredible, ...
1	3	1	cat, ninja, climbing, ve, photo, taken, best, ...
2	3	2	restaurant, play, came, eating, ltitle, kitty,...
3	4	0	google, translate, feedback, map, incredible, ...
4	4	1	cat, ninja, climbing, ve, photo, taken, best, ...
5	4	2	restaurant, play, came, eating, ltitle, kitty,...
6	4	3	key, promoter, chrome, extension, google, ve, ...
7	5	0	google, translate, feedback, map, incredible, ...
8	5	1	cat, ninja, climbing, ve, photo, taken, best, ...
9	5	2	restaurant, play, came, eating, ltitle, kitty,...
10	5	3	key, promoter, chrome, extension, google, ve, ...
11	5	4	merley, squooshy, bets, belly, kitten, ve, tra...
12	6	0	google, translate, feedback, map, incredible, ...
13	6	1	cat, ninja, climbing, ve, photo, taken, best, ...
14	6	2	restaurant, play, came, eating, ltitle, kitty,...
15	6	3	key, promoter, chrome, extension, google, ve, ...
16	6	4	merley, squooshy, bets, belly, kitten, ve, tra...
17	6	5	smiley, tab, 100, face, open, google, translat...
18	7	0	translate, app, incredible, google, ve, taken,...
19	7	1	cat, ninja, climbing, ve, photo, taken, best, ...
20	7	2	restaurant, play, came, eating, ltitle, kitty,...
21	7	3	key, promoter, chrome, extension, google, ve, ...
22	7	4	merley, squooshy, bets, belly, kitten, ve, tra...
23	7	5	smiley, tab, 100, face, open, google, translat...

24	7	6	impressed, map, feedback, google, translate, v...
25	8	0	translate, app, incredible, google, ve, taken,...
26	8	1	climbing, ninja, cat, ve, translate, taken, re...
27	8	2	restaurant, play, came, eating, ltitle, kitty,...
28	8	3	merley, squooshy, bets, belly, kitten, ve, tra...
29	8	4	smiley, tab, 100, face, open, google, translat...
30	8	5	impressed, map, feedback, google, translate, v...
31	8	6	ve, taken, best, photo, cat, translate, smiley...
32	8	7	key, promoter, chrome, extension, google, ve, ...